

# DAVID WEBSTER

## AI Systems Builder | Agent Infrastructure, Applied ML & 0-to-1 Products

Amsterdam, NL | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

I turn ambiguous technical problems into working systems. Over 12 years I have owned architecture, critical-path code, model training and technical delivery across biometric identity, applied ML, distributed data platforms and agent infrastructure. My current focus is the boundary that determines what autonomous systems can read, call and change.

### Products & Company Thesis

- **SHIPPED - llm-input-hardening:** Published a Rust-backed Python package that detects and normalises hostile Unicode before text reaches prompts, retrieval, logs or tool arguments. Ships with fuzzing, telemetry, reproducible benchmarks and a 154-case labelled evaluation corpus. [PyPI](#)
- **WORKING PREVIEW - Blacksmith / agent-api-hardening:** Turns OpenAPI contracts into least-capability SDK and MCP surfaces. Unreviewed writes and destructive, financial or administrative operations are removed from the agent-facing interface by construction; verified with 86 tests and an explicit quality corpus.
- **POC - Greenlight:** Built a decision, approval and audit layer for AI and human actions with policy evaluation, multi-party approval, idempotent mutations, signed webhooks, audit export and JavaScript/React SDKs. [Demo](#)

### Technical Depth

**Agent systems:** control and data planes, platform APIs, MCP, agent-safe tool surfaces, policy boundaries, tracing and evaluation

**Production ML:** trained computer-vision and NLP models and built the process engines, data paths and deployment platforms around them

**R&D:** AWS P5/H100, ParallelCluster and Slurm; CNN, LSTM and NLP training with TensorFlow/PyTorch; CUDA; autoencoders, GANs and quantum prototypes

**Software & systems:** Python, Rust, Java, C++, TypeScript | Linux, AWS, Kubernetes, Terraform, Kafka, Flink, Spark, EMR, Airflow and SageMaker

### 0-to-1 & Production Proof

- **0-to-1 delivery** - Built BuzBrain's first generative-AI product in under two weeks; solo-built and launched InvestSure's production claims platform in under two months; rebuilt and delivered Standard Bank's ML platform in three months.
- **Models plus systems** - Led technical delivery of Absa's ML brand-audit system, building its AWS process engine and training TensorFlow models for logo and text detection. It analysed 5,000 records in 30 minutes and helped audit 40,000 artefacts across 300 systems. [Case study](#)
- **Production from the metal up** - Built and shipped Java web services and C++ biometric backends used across customer deployments in Guinea, Ghana, Namibia and Botswana, including on-site infrastructure rollout and production support.
- **Commercial and technical ownership** - Brought TMNL into Luminis as a strategic client and led architecture across platform, MLOps and privacy-enhancing technology workstreams; earlier ran Synthesis engagements from model training through delivery.

### Current Work

#### Backbase | Principal Solution Architect - Agentic Platform

Sep 2025 - Present

- Build Backbase's shared agentic platform across application and infrastructure layers, including APIs behind its control and data planes.
- Work with product and customer teams on use cases and production architectures, turning repeated requirements into reusable platform capabilities. Agentic functionality is live in production.

## Earlier Experience

### Luminis | Senior Cloud Architect

Oct 2024 - Sep 2025

- Brought TMNL into Luminis as a strategic client and continued leading architecture across platform engineering, MLOps and PET workstreams.
- Built BuzBrain's generative-AI product in under two weeks with AWS Bedrock, Bubble.io and Flowise; helped build ING's real-time Kafka/Flink platform and Gradyent's digital-twin infrastructure.

### TMNL | Tech Lead Platform Engineering, L6

Oct 2021 - Sep 2024

- Built an analytics and ML platform across five Dutch banks, using a cross-account catalogue, schema registry and access boundaries so sensitive workloads could run without duplicating source data.
- Evolved workloads from Kafka, Flink and SageMaker to Spark, EMR and Airflow; led adoption across data and MLOps teams and reduced infrastructure cost by \$1.5M.
- Supported R&D with TNO and Roseman Labs across MPC, homomorphic encryption, federated learning and quantum-resistant approaches.

### Schuberg Philis | AWS Technical Practice Lead / APN Ambassador

Jan 2020 - Oct 2021

- Built LeasePlan's global AWS platform and completed its on-premises migration in under six months, including reusable Terraform/Kubernetes migration factories.
- Built Lab271 R&D prototypes across DeepRacer, autoencoders, GANs, deepfakes, Jetson and quantum games using SageMaker, TensorFlow, PyTorch, Braket and Qiskit.

### AAme Solutions / ABN AMRO | Software / Cloud Engineer

May 2019 - Dec 2019

- Migrated IBM Java services to AWS Lambda, built a real-time maintenance tool, introduced BFF patterns across digital channels and contributed to early personalisation architecture in a 1B+ annual-login environment.

### Synthesis | Software / Cloud Engineer

Nov 2016 - May 2019

- Worked directly with the Chief Disruption Officer and Head of Cloud Consulting on 0-to-1 engagements, owning architecture, core implementation, model training and technical customer delivery.
- Led Absa's ML brand-audit delivery; took over and delivered Standard Bank's personalisation platform in three months for a bank with 19M+ customers.
- Solo-built and launched InvestSure's serverless claims platform in under two months; claims processed in under ten seconds at less than \$500 per month.

### Paycode | Software Engineer

Aug 2014 - Nov 2016

- Built and shipped Java web services and C++ biometric backends for identity, remittance, KYC and AML products used in customer deployments across four African countries; handled on-site rollout and production support.

## Education & Credentials

BSc Computer Science | University of South Africa | 2011 - 2015

13 AWS certifications earned | 5 Kubernetes credentials | TensorFlow Developer Certificate | 2 NVIDIA CUDA certificates | 2

DeepLearning.AI course certificates